

Refusal Behavior in Large Language Models: A Nonlinear Perspective

Fabian Hildebrandt
CCN Group
Pattern Recog. Lab.
FAU Erlangen-Nürnberg
Erlangen, Germany
fabian.hildebrandt@fau.de

Andreas Maier
Pattern Recog. Lab.
FAU Erlangen-Nürnberg
Erlangen, Germany
andreas.maier@fau.de

Patrick Krauss*
CCN Group
Pattern Recog. Lab.
FAU Erlangen-Nürnberg
Erlangen, Germany
patrick.krauss@fau.de

Achim Schilling*
Neuroscience Lab,
University Hospital Erlangen,
CCN Group
Pattern Recog. Lab.
FAU Erlangen-Nürnberg
Erlangen, Germany
achim.schilling@fau.de

Abstract—Refusal behavior in large language models (LLMs) enables them to decline responding to harmful, unethical, or inappropriate prompts, ensuring alignment with ethical standards. This paper investigates refusal behavior across six LLMs from three architectural families. We challenge the assumption of refusal as a linear phenomenon by employing dimensionality reduction techniques, including PCA, t-SNE, and UMAP. Our results reveal that refusal mechanisms exhibit nonlinear, multidimensional characteristics that vary by model architecture and layer. These findings highlight the need for nonlinear interpretability to improve alignment research and inform safer AI deployment strategies.

Index Terms—refusal, mechanistic interpretability, LLM, AI alignment, neuroscience-inspired AI (neuroAI), explainable AI (XAI)

INTRODUCTION

Morality and impulse control are fundamental aspects of human behavior, enabling ethical decision-making, effective social interactions, and the maintenance of personal and societal relationships. The amygdala and the ventromedial prefrontal cortex play critical roles in moral behavior [1]. The amygdala associates harmful actions with negative emotions, while the ventromedial prefrontal cortex governs decision-making and self-control to prevent such actions. In psychopathy, these regions exhibit reduced activity, leading to an increased propensity for manipulation, impulsivity, and unethical decisions without remorse [2]. Impulse control, a key aspect of self-regulation housed in the prefrontal cortex, can be disrupted by various factors such as ADHD, substance use disorders, psychological conditions, stress, and brain injuries [3].

The human brain is often described as a prediction machine, continuously processing sensory information to guide decision-making. Similarly, Large Language Models (LLMs) are designed to predict the next word or sequence, enabling them to perform complex language tasks. Like the human brain, LLMs are tasked with handling nuanced decisions, such as refusing harmful instructions, yet their internal decision-making processes often remain opaque. Efforts to align LLMs with ethical principles have focused on feature search, which seeks to interpret model activations and uncover the mecha-

nisms governing their behavior. The superposition hypothesis posits that multiple concepts or features are simultaneously represented within the same neural activation space [4], [5]. Consequently, individual neurons in LLMs often exhibit polysemantic behavior, encoding multiple distinct meanings rather than a one-to-one mapping. Achieving monosemantic representations—where neurons encode a single, unambiguous feature—is a critical goal for improving interpretability. Sparse autoencoders have been employed to identify such features by inflating the hidden space and enforcing sparsity [6]–[9]. This approach enables the isolation of specific features for steering model behavior [10]–[12].

Recent studies suggest that refusal behavior in LLMs is mediated by a single linear subspace in the activation space. This feature can be manipulated to either disable or enforce refusal behavior across a range of open-source models, including those with up to 72B parameters [13]. Techniques like the difference-in-means method [14] and weight orthogonalization have been used to isolate and modify this subspace, resulting in models that either lose or gain the ability to refuse harmful instructions. However, emerging evidence challenges the assumption that refusal behavior resides in a linear subspace, pointing instead to a multidimensional and nonlinear nature [15].

In this work, we examine refusal behavior across six LLMs spanning three model families. By analyzing intermediate layer activations through both linear (PCA) and nonlinear (t-SNE, UMAP) dimensionality reduction techniques, we reveal that refusal behavior is a universal but architecture-specific phenomenon. Our findings show that refusal mechanisms are more complex and nonlinear than previously assumed, with distinct sub-clusters emerging in the activation space. These insights contribute to a deeper understanding of refusal behavior and its implications for aligning LLMs with ethical and safety standards.

METHODOLOGY

Behavioral studies and neural correlates

In this study, we investigate the refusal behavior of large language models (LLMs) in response to harmful and harm-

less instructions, drawing parallels to behavioral studies that examine brain activity linked to cognitive functions. Our methodology involves designing a task that isolates the refusal mechanisms in LLMs, similar to decision-making tasks in behavioral research that require empathy and sensitivity. Simultaneously, we track the LLM activations to identify model activations associated with the refusal behavior. Additionally, the origin of these responses is localized and the embeddings are further investigated.

Datasets

Two distinct datasets are used. The first dataset D_{harmless} contains harmless instructions from the ALPACA dataset [16] repackaged and published as a Hugging Face dataset [17]. The ALPACA dataset is a collection of 52000 harmless instruction-following prompts designed to fine-tune large language models for more effective and reliable task completion. The second dataset D_{harmful} contains harmful instructions that originate from the *LLM Attacks* dataset [18], which was designed for adversarial attacks against aligned language models. The instructions are trying to induce harmful behavior by the LLMs. Again, a repackaged Hugging Face dataset is used [19].

Models

To investigate the universality of the refusal behavior, we evaluated a set of six large language models (LLMs) from three model families. All models are fine-tuned for instruction following. The selection of different model families and varying parameter sizes allows to assess whether the observed effects are consistent and generalizable. The model families represent different alignment types from preference optimization to alignment by fine-tuning. The specific models used in this study are detailed in Table I.

Extraction of the refusal activations

The methodology for extracting activations related to the refusal mechanism follows the general approach outlined in [13] and [23]. First, we load the model using the *TranformerLens* library [24]. The library allows to cache any internal activation in the model for mechanistic interpretability research. Next, two equal-sized subsets of prompts from the two datasets D_{harmful} and D_{harmless} containing harmful and harmless instructions are loaded using the Hugging Face *Datasets* library. Using a chat-style generation template, we run the model on both datasets while caching the activations for each layer. To identify the refusal-related feature direction, we compute the difference-in-means by subtracting the mean activations of harmless prompts from those of harmful prompts at each token position. Finally, we store the residual activations after the self-attention layers and the multilayer perceptron (MLP) layers at the last token position, for further analysis and manipulation.

Difference-in-means. The difference-in-means technique [14] can be used to identify the refusal direction in the model’s activations as shown in previous works [13], [23].

For each layer $l \in [L]$ the mean activation $\mu_{\text{harmful}}^{(l)}$ for harmful prompts from D_{harmful} and $\mu_{\text{harmless}}^{(l)}$ for harmless prompts from D_{harmless} is calculated:

$$\begin{aligned}\mu_{\text{harmful}}^{(l)} &= \frac{1}{|D_{\text{harmful}}|} \sum_{i \in D_{\text{harmful}}} y^{(l)}(i), \\ \mu_{\text{harmless}}^{(l)} &= \frac{1}{|D_{\text{harmless}}|} \sum_{i \in D_{\text{harmless}}} y^{(l)}(i).\end{aligned}\tag{1}$$

where $y_i^{(l)}(t)$ represents the activation at the last token position in layer l for instruction i . The difference-in-means direction $d_{\text{refusal}}^{(l)}$ is then computed as:

$$d_{\text{refusal}}^{(l)} = \mu_{\text{harmful}}^{(l)} - \mu_{\text{harmless}}^{(l)}.\tag{2}$$

The vector $d_{\text{refusal}}^{(l)}$ captures both the direction and the magnitude of the refusal feature in the activations.

Dimensionality reduction

Principal Component Analysis (PCA). PCA is a widely used linear dimensionality reduction technique that transforms an unlabeled dataset into a new coordinate system where the greatest variance by any projection of the data lies on the first principal component, the second greatest variance on the second component, and so on [25]. This method identifies orthogonal axes that maximize variance, thereby simplifying high-dimensional data while preserving the maximum variability. PCA is commonly applied for feature extraction and the feature visualization.

t-Distributed Stochastic Neighbor Embedding (t-SNE). t-SNE is a nonlinear dimensionality reduction technique commonly used for visualizing high-dimensional data in two or three dimensions. It models the pairwise similarities between points in the original space and seeks to preserve these relationships in the reduced space by minimizing a divergence between two probability distributions. Unlike linear methods like PCA, t-SNE is effective at capturing local structure and revealing clusters or patterns within complex datasets [26].

Uniform Manifold Approximation and Projection (UMAP). UMAP is a dimensionality reduction technique designed for visualization and the general nonlinear embedding of high-dimensional data. Based on manifold learning and topological data analysis, UMAP constructs a high-dimensional graph of data points and optimizes a low-dimensional projection to preserve both local and global structures. Compared to PCA, UMAP captures complex, nonlinear relationships in data, making it more effective for revealing meaningful clusters. Unlike t-SNE, UMAP offers faster computation, better scalability, and more interpretable low-dimensional embeddings while maintaining the ability to preserve local neighborhood information [27].

Activation separability metric

Generalized Discrimination Value (GDV). To quantify the degree of clustering, we used the GDV as published

TABLE I
COMPARISON OF MODEL SPECIFICATIONS

Model Name	Model Size	Number of Layers	Hidden Space Dimensions	Model Origin
Llama-3.2-1B-Instruct	1.1B	16	2048	Meta AI [20]
Llama-3.2-3B-Instruct	3.2B	28	3072	Meta AI [20]
Bloom-560M-Instruct	302M	24	1024	BigScience [21]
Bloom-3B-Instruct	2.4B	30	2560	BigScience [21]
Qwen2-0.5B-Instruct	391M	24	896	Alibaba Group [22]
Qwen2-1.5B-Instruct	1.4B	28	1536	Alibaba Group [22]

and explained in detail in [28]. The GDV provides an objective measure of how well the hidden layer activations cluster according to the ASC types, offering insights into the model’s internal representations. Briefly, we consider N points $\mathbf{x}_{n=1..N} = (x_{n,1}, \dots, x_{n,D})$, distributed within D -dimensional space. A label l_n assigns each point to one of L distinct classes $C_{l=1..L}$. In order to become invariant against scaling and translation, each dimension is separately z-scored and, for later convenience, multiplied with $\frac{1}{2}$:

$$s_{n,d} = \frac{1}{2} \cdot \frac{x_{n,d} - \mu_d}{\sigma_d}. \quad (3)$$

Here, $\mu_d = \frac{1}{N} \sum_{n=1}^N x_{n,d}$ denotes the mean,

and $\sigma_d = \sqrt{\frac{1}{N} \sum_{n=1}^N (x_{n,d} - \mu_d)^2}$ the standard deviation of dimension d .

Based on the re-scaled data points $\mathbf{s}_n = (s_{n,1}, \dots, s_{n,D})$, we calculate the *mean intra-class distances* for each class C_l

$$\bar{d}(C_l) = \frac{2}{N_l(N_l-1)} \sum_{i=1}^{N_l-1} \sum_{j=i+1}^{N_l} d(\mathbf{s}_i^{(l)}, \mathbf{s}_j^{(l)}), \quad (4)$$

and the *mean inter-class distances* for each pair of classes C_l and C_m

$$\bar{d}(C_l, C_m) = \frac{1}{N_l N_m} \sum_{i=1}^{N_l} \sum_{j=1}^{N_m} d(\mathbf{s}_i^{(l)}, \mathbf{s}_j^{(m)}). \quad (5)$$

Here, N_k is the number of points in class k , and $\mathbf{s}_i^{(k)}$ is the i^{th} point of class k . The quantity $d(\mathbf{a}, \mathbf{b})$ is the euclidean distance between $\mathbf{a}$ and $\mathbf{b}$. Finally, the Generalized Discrimination Value (GDV) is calculated from the mean intra-class and inter-class distances as follows:

$$\text{GDV} = \frac{1}{\sqrt{D}} \left[\frac{1}{L} \sum_{l=1}^L \bar{d}(C_l) - \frac{2}{L(L-1)} \sum_{l=1}^{L-1} \sum_{m=l+1}^L \bar{d}(C_l, C_m) \right] \quad (6)$$

whereas the factor $\frac{1}{\sqrt{D}}$ is introduced for dimensionality invariance of the GDV with D as the number of dimensions.

Note that the GDV is invariant with respect to a global scaling or shifting of the data (due to the z-scoring), and also invariant with respect to a permutation of the components in the N -dimensional data vectors (because the euclidean distance

measure has this symmetry). The GDV is zero for completely overlapping, non-separated clusters, and it becomes more negative as the separation increases. A GDV of -1 signifies already a very strong separation and perfect clustering.

RESULTS

Refusal is a universal feature

Refusal behavior is consistently observed across all six tested models, with a distinct separation between harmful and harmless instructions in the dimensionality-reduced residual activations. This separation is evident across model families and sizes. Figure 1 illustrates the refusal feature in three models at different layers. Table II summarizes the lowest GDV values, indicating the best separability between harmful and harmless instructions, along with the corresponding model layers and dimensionality-reduction techniques.

TABLE II
COMPARISON OF SEPARABILITY ACROSS MODELS AND METHODS

Model	Method	GDV	Layer
Llama-3.2-1B-Instruct	PCA	-0.45	12/16 (ATTENTION)
Llama-3.2-1B-Instruct	t-SNE	-0.62	9/16 (MLP)
Llama-3.2-1B-Instruct	UMAP	-0.69	16/16 (MLP)
Llama-3.2-3B-Instruct	PCA	-0.51	24/28 (ATTENTION)
Llama-3.2-3B-Instruct	t-SNE	-0.60	11/28 (MLP)
Llama-3.2-3B-Instruct	UMAP	-0.80	15/28 (ATTENTION)
Qwen2-0.5B-Instruct	PCA	-0.46	1/24 (ATTENTION)
Qwen2-0.5B-Instruct	t-SNE	-0.73	5/24 (ATTENTION)
Qwen2-0.5B-Instruct	UMAP	-0.87	5/24 (MLP)
Qwen2-1.5B-Instruct	PCA	-0.44	4/28 (ATTENTION)
Qwen2-1.5B-Instruct	t-SNE	-0.71	2/28 (MLP)
Qwen2-1.5B-Instruct	UMAP	-0.89	18/28 (ATTENTION)
bloom-3b	PCA	-0.40	6/30 (MLP)
bloom-3b	t-SNE	-0.63	28/30 (ATTENTION)
bloom-3b	UMAP	-0.66	25/30 (MLP)
bloom-560m	PCA	-0.41	5/24 (MLP)
bloom-560m	t-SNE	-0.60	23/24 (ATTENTION)
bloom-560m	UMAP	-0.65	22/24 (MLP)

Refusal is a nonlinear feature

Our analysis demonstrates that refusal behavior in LLMs extends beyond a simple one-dimensional linear subspace, revealing a complex, nonlinear structure. Using PCA, UMAP, and t-SNE, we visualized dimensionality-reduced activations for harmful and harmless instructions across multiple layers of six LLMs. While PCA, a linear technique, captures variance through linear distances, UMAP and t-SNE effectively